

Proposition 1 (The Implicit Function Theorem). *(What is F?)* $F : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$, If $F(a, b) = 0$ and $F(x, y)$ is continuously differentiable on some open disk with center (a, b) , then, if

$$\frac{\partial F}{\partial y}(a, b) \neq 0$$

there exists an $h > 0$ and a unique function $\phi(x)$ continuously differentiable *(What can we say about ϕ ?)* defined for $|x - a| < h$ such that $\phi(a) = b$ and $F(x, \phi(x)) = 0$ for $|x - a| < h$.

Proposition 2 (The Actual Pasting Lemma). Let X and Y be metric spaces. Let $(A, B) \in \text{Closed}(X)^2$ and assume further that $X = A \cup B$. Assume that we have two functions $f : A \rightarrow Y$ and $g : B \rightarrow Y$. Assume that $f(x) = g(x) \quad \forall x \in A \cap B$. Then we can define a new function $h : X \rightarrow Y$ by:

$$h(x) = \begin{cases} f(x) & \text{if } x \in A, \\ g(x) & \text{if } x \in B \end{cases}$$

(if $x \in A \cap B$ then we have no ambiguity by assumption). **Claim:** If both f and g are continuous, then h is continuous.

Definition 1. Let $P \in \mathbb{R}[x, y]$ be a squarefree polynomial, with $\deg_y(P) > 0$. We denote by $\text{proj}(P)$ the set of points $x \in \mathbb{R}$ such that either $\text{Disc}_y(P)(x) = 0$ or $\text{lc}_y(P)(x) = 0$.

Exercise 1. Let $P \in \mathbb{R}[x, y]$ be a squarefree polynomial, with $\deg_y(P) > 0$. Suppose that $\#\text{proj}(P) < \infty$. Let $x_1, x_2, \dots, x_s$ be the elements of $\text{proj}(P)$ in ascending order, and let $x_0 = -\infty$ and $x_{s+1} = +\infty$. Show that for all $i = 0, 1, \dots, s$, the following properties hold:

1. For all $x \in (x_i, x_{i+1})$, $\{y \in \mathbb{R} : P(x, y) = 0\}$ is finite.
2. For all $(x, y) \in (x_i, x_{i+1}) \times \mathbb{R}$ such that $P(x, y) = 0$, there exists an open neighborhood I_x of x and a unique function $g \in \mathcal{C}^1(I_x)$ such that $g(x) = y$ and $\forall x' \in I_x$, $P(x', g(x')) = 0$.

Let $x \in (x_i, x_{i+1})$, $t = \#\{y \in \mathbb{R} : P(x, y) = 0\}$, and let $y_1, y_2, \dots, y_t$ be the roots of $P(x, y)$ in ascending order. Show that the following properties hold:

3. For all $j = 1, 2, \dots, t$, let g_j be the local function given by the previous lemma at (x, y_j) . Show that g_j extends uniquely to a continuous function $h_j : (x_i, x_{i+1}) \rightarrow \mathbb{R}$ such that $h_j(x) = y_j$ and for all $x' \in (x_i, x_{i+1})$, $P(x', h_j(x')) = 0$.
4. In addition, $h_1(x') < h_2(x') < \dots < h_t(x')$ for all $x' \in (x_i, x_{i+1})$.
5. For all $x' \in (x_i, x_{i+1})$, $\#\{y \in \mathbb{R} : P(x', y) = 0\} \geq t$.

Finally, show that the number of roots of $P(x, y)$ in $\mathbb{R}$ is constant for all $x \in (x_i, x_{i+1})$.

- Proof.* 1. For property 1, We know that $\forall x \in (x_i, x_{i+1}), lc_y(P)(x) \neq 0$, so the polynomial in y $P(x, y)$ isn't the zero polynomial. It's impossible that $P(x, y)$ has an infinite number of roots. In conclusion, $\{y \in \mathbb{R} : P(x, y) = 0\}$ is finite.
2. For property 2, $\forall (x, y) \in (x_i, x_{i+1}) \times \mathbb{R}$, we have $P(x, y) = 0$ and $Disc_y(P)(x) \neq 0$. P is polynomial function, so it's continuously differentiable. According to the Implicit Function Theorem, there exists an open neighborhood I_x of x , and a unique function $g \in \mathcal{C}^1(I_x)$, such that $g(x) = y$ and $\forall x' \in I_x, P(x', g(x')) = 0$.
3. For property 3, We firstly show that locally we can get a unique continuous function. Suppose two points $(a, b), (c, d)$, obssessing the functions $g_{j,a} : I_a \rightarrow \mathbb{R}, g_{j,c} : I_c \rightarrow \mathbb{R}$ satisfying that $a, c \in (x_i, x_{i+1})$ and the open neighborhoods $A = I_a \cap I_c \neq \emptyset$. For all $e \in A$, we have $P(e, g_{j,a}(e)) = P(e, g_{j,c}(e)) = 0$, locally it's evident that $g_{j,a}(e) = g_{j,c}(e)$. According to the Pasting Lemma, we can construct a function $h_{j,a,c} : I_a \cup I_c \rightarrow \mathbb{R}$, such that $\forall x' \in I_a \cup I_c, P(x', h_{j,a,c}(x')) = 0$. We know that $g_{j,a}, g_{j,c}$ are unique, so the function $h_{j,a,c}$ is unique as well. Now $I \in (x_i, x_{i+1})$ is the biggest interval we can obtain by Using Pasting Lemma. Suppose that $I \neq (x_i, x_{i+1})$, then we can find an index $a \in (x_i, x_{i+1}) \setminus I$, satisfying that $I_a \cap I \neq \emptyset$. By using the Pasting Lemma, we can construct a bigger interval $I \cup I_a \in (a_i, a_{i+1})$, contradiction. So $I = (x_i, x_{i+1})$. Like the local procedure, we can get the h_j
4. For property 4,

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